A COMMUTATIVE-UP-TO-HOMOTOPY SQUARE INDUCES A MORPHISM OF THE CONES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-functorial-cone (source lines 643-662); the proof environment begins two lines below it and is its own proof, closing at source line 680.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the cone and the homotopy category are the author's own surrounding development in the same chapter.

Editorial note. The statement contains an Xy-pic square and the proof displays one block matrix; both are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: a homotopy commutative square of dg modules induces a morphism between the cones, giving a morphism of triangles.

Allowed standard prerequisites (editorial).

- Cone of a morphism of dg modules. The cone C(f) of f : X -> Y is the dg module Y direct sum X[1] carrying the inclusion i, the projection p and the differential built from d_Y, f and the shifted d_X.. The maps i_1, p_1, i_2, p_2 and the differentials of the two cones are what the constructed c must be compatible with (source lines 655-658, 672-680).
- Homotopy between morphisms of dg modules. A homotopy between two morphisms of dg modules is a graded map h of degree minus one such that their difference equals d h plus h d.. The homotopy h between f_2 a and b f_1 becomes the off-diagonal entry of the matrix defining c and is what makes c a morphism of dg modules (source lines 666-672).
- Morphism of triangles in the homotopy category. A morphism between two distinguished triangles is a triple of morphisms making the two relevant squares commute in the homotopy category.. The conclusion is that the triple (a, b, c) is such a morphism (source lines 674-680).

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Lemma 0.1. *Let (A, d) be a differential graded algebra. Suppose that*

$$\begin{array}{ccc} K_1 & \xrightarrow[f_1]{} & L_1 \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ K_2 & \xrightarrow{f_2} & L_2 \end{array}$$

is a diagram of homomorphisms of differential graded A-modules which is commutative up to homotopy. Then there exists a morphism $c : C(f_1) \to C(f_2)$ which gives rise to a morphism of triangles

$$(a, b, c) : (K_1, L_1, C(f_1), f_1, i_1, p_1) \to (K_1, L_1, C(f_1), f_2, i_2, p_2)$$

in $K(Mod_{(A,d)})$.

Proof. Let $h : K_1 \to L_2$ be a homotopy between $f_2 \circ a$ and $b \circ f_1$. Define c by the matrix

$$c = \begin{pmatrix} b & h \\ 0 & a \end{pmatrix} : L_1 \oplus K_1 \to L_2 \oplus K_2$$

A matrix computation show that c is a morphism of differential graded modules. It is trivial that $c \circ i_1 = i_2 \circ b$, and it is trivial also to check that $p_2 \circ c = a \circ p_1$. $\square$